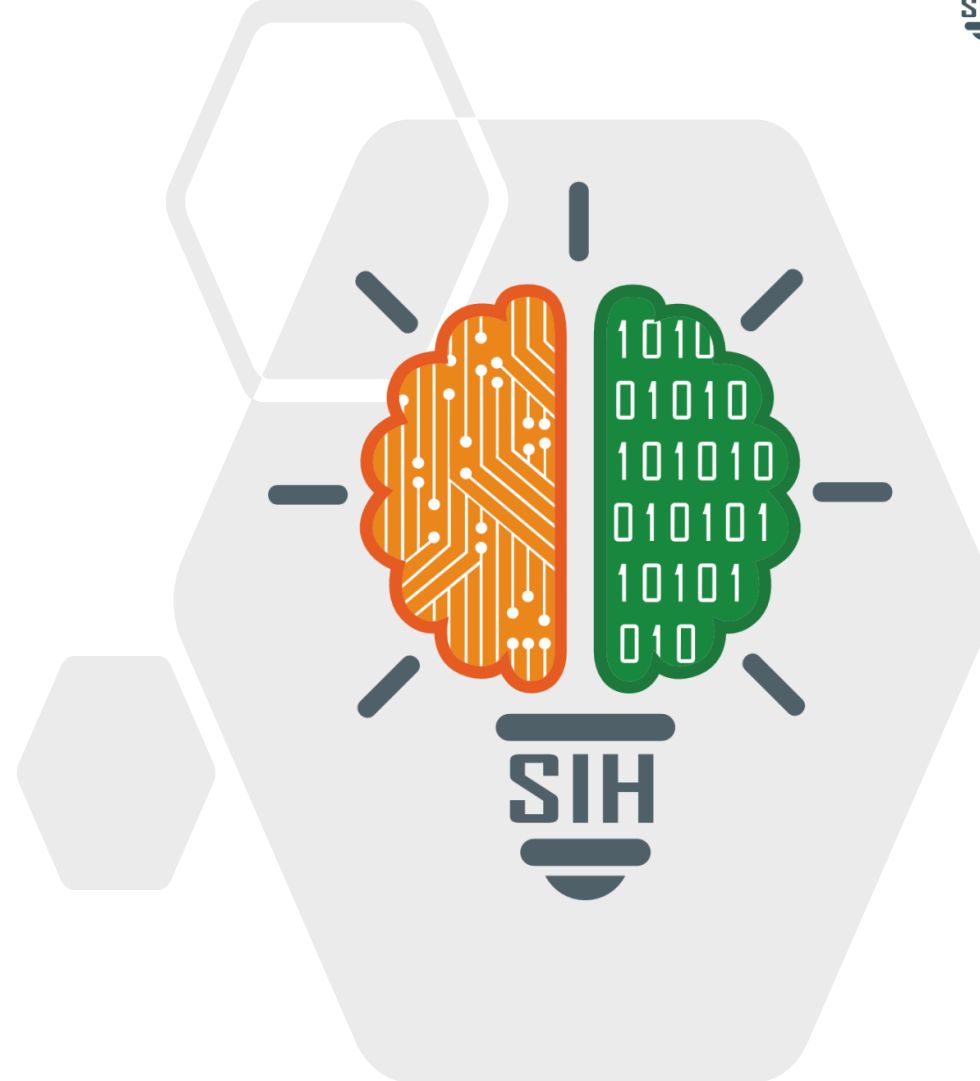


SMART INDIA HACKATHON 2026



- **Problem Statement ID : SIH26175**
- **Problem Statement Title : DepthWizard - Single-View Height Estimation and 3D Flythrough**
- **Theme : Disaster Management**
- **PS Category : Software**
- **Team ID : 129655**
- **Team Name : Dev Up**



THE PROBLEM

"[Stereo, LiDAR and InSAR] can be cost-prohibitive, dependent on specific sensor availability, and computationally intensive."
— ISRO, problem statement SIH26175

- 01 Archives with no height**
Stereo and LiDAR cannot be flown into the past. Decades of Cartosat archive carry no height.
- 02 Relative depth, not metres**
Depth models learn from ground-level photos. From orbit they give shape, but no metric scale.
- 03 No confidence, no product**
A bare height hides where it is wrong, and a checkpoint is not a scene an analyst can open.

WHO FEELS IT

SAC analysts · disaster responders · planners · foresters

OUR SOLUTION

 **One image to a metric DSM**
Depth Anything V2, fine-tuned on airborne-LiDAR heights. GeoTIFF out, metadata or not.

 **Validation inside the product**
RMSE, MAE and correlation against LiDAR — per terrain class, not one flattering average.

 **Confidence, not just height**
A per-pixel sigma beside every height, so the model says where it is unsure.

 **Absolute heights, two ways**
Copernicus GLO-30 for any GeoTIFF, or ground control points via a power-law fit.

 **A navigable 3D scene**
three.js flythrough with measure, flood and drag-to-compare-against-LiDAR tools.

 **Deployable, not a demo**
No GPU needed. The viewer is one HTML file that opens from disk, offline.

THE GAP WE CLOSE

The single-view papers we build on stop at weights and a score. **We ship metric height, calibrated confidence and a no-GPU 3D viewer** — compared row by row on slide 6.

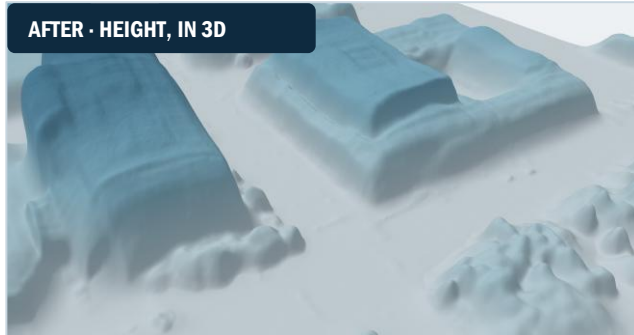
THE IDEA, RUNNING: ONE ARCHIVED IMAGE IN, A MEASURABLE SURFACE OUT — CAPTURES OF OUR PROTOTYPE, NOT MOCK-UPS

BEFORE · INPUT



One Omaha tile, as flown.

AFTER · HEIGHT, IN 3D



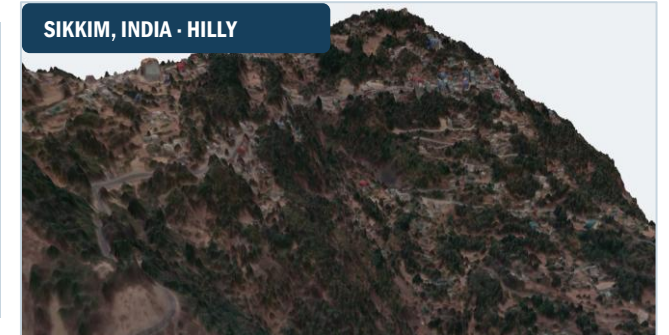
Same tile: roofs, tree crowns and kerbs at 0.3 m.

AFTER · CONFIDENCE



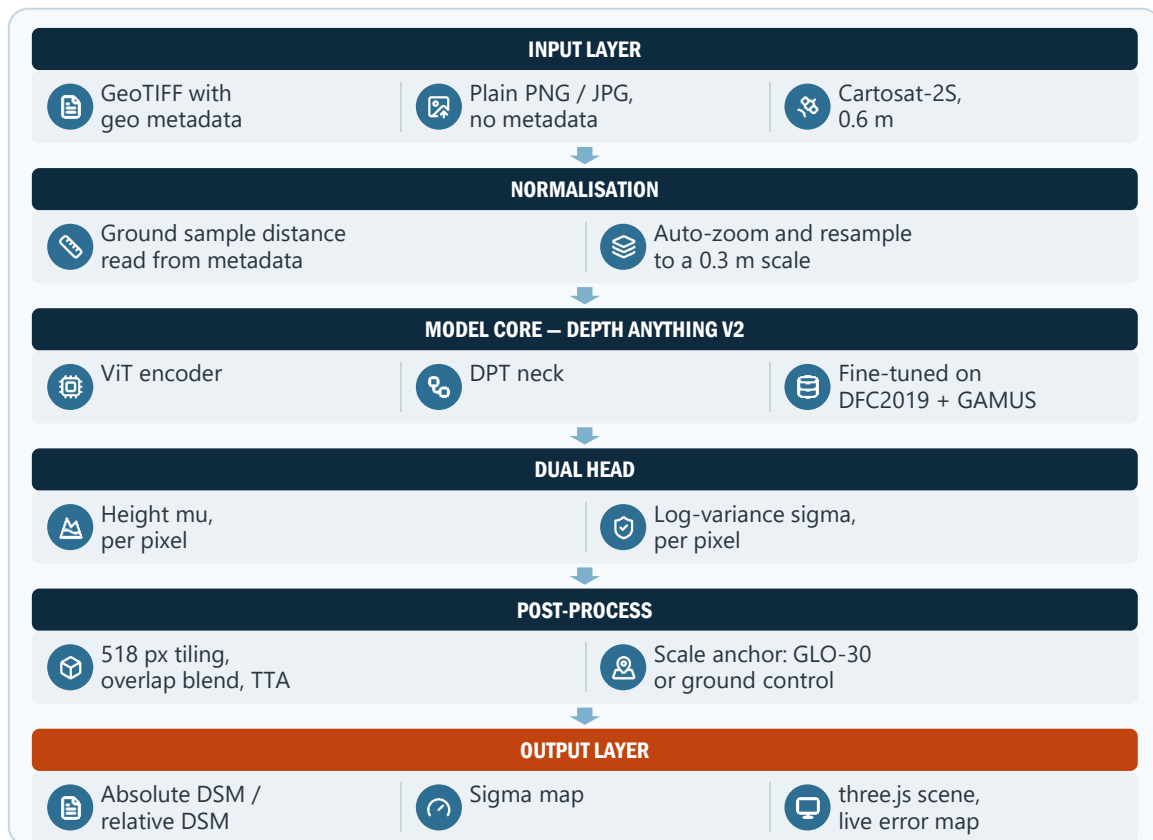
Where the estimate is weak, the sigma says so.

SIKKIM, INDIA · HILLY



967 m of relief across 2 km. Imagery © Maxar, CC BY-NC 4.0

ARCHITECTURE — ONE IMAGE TO A CHECKED SURFACE



HOW IT IS VALIDATED

Split region-disjoint, not random — adjacent tiles share buildings.
Scored on whole held-out tiles against airborne LiDAR, never on crops.
Reported per urban / sparse / forested / mixed, and per height band.
Held back no trained model is scored on the test split until the end.

HOW IT RUNS — HARDWARE, API, DATA

- 1 Trained on one consumer GPU**
RTX 3060, 12 GB. Backbone: Depth Anything V2 Small, Apache 2.0, 24.8 M parameters.
- 2 Exported for CPU**
ONNX, int8-quantised to 36.8 MB. Inference needs no GPU at all.
- 3 Served over HTTP**
POST /api/upload takes TIF, PNG or JPG; /api/result returns the DSM. Live on a two-core CPU host.
- 4 Files, not a database**
GeoTIFF DSM and sigma, float32 scene tiles. The viewer is one 15 MB HTML file that opens from disk.

DESIGN CHOICES, AND WHY

- Metres learned from LiDAR**
Fine-tuning on airborne-LiDAR heights makes the output metric directly; a DEM supplies the ground under it.
- Uncertainty as an output**
A twin head predicts sigma beside every height (heteroscedastic NLL with beta-NLL), so doubt is visible.
- Built for ISRO's 0.6 m**
Auto-zoom reads a GeoTIFF's resolution and resamples Cartosat-2S to the scale the model learned.
- Chosen by measurement**
Regression, binned and head-tail-cut heads were all trained and scored; the simplest held its own, so it ships.

TECHNOLOGY STACK — EVERY DEPENDENCY FREE AND OPEN



**TECHNICALLY PROVEN**

Built and measured: slide 7.

**OPERATIONALLY SIMPLE**

One image, CPU only, works offline.

**ECONOMICALLY VIABLE**

₹0 licences, runs on existing PCs.

**ALREADY WORKING**

All three ISRO milestones are built.

THE HARD PROBLEMS, AND HOW EACH IS HANDLED**PROBLEM** ISRO judges on 0.6 m Cartosat-2S; open LiDAR training imagery is 0.3 m**HANDLED** Auto-zoom reads the GeoTIFF's pixel size and resamples to the scale the model learned. Its cost at 0.6 m is measured on the shipped model (slide 7); fine-tuning on 0.6 m-degraded tiles is next.**PROBLEM** Cartosat-2S colour is not the aerial colour the model learned from**HANDLED** Training jitters brightness, contrast and gamma to mimic sensor and haze; any band count loads. The cost on real Cartosat colour is not yet measured.**PROBLEM** One image has no stereo parallax, so which heights can be trusted?**HANDLED** Every pixel ships with an uncertainty (σ), calibrated against LiDAR. Unsure pixels are flagged for the analyst, not hidden.**PROBLEM** An absolute DSM needs ground elevation a single image cannot give**HANDLED** Copernicus GLO-30 terrain plus our above-ground heights, on by default. GLO-30's own error (< 4 m, LE90) is inherited, and stated.**PROBLEM** Field offices have no GPU and often no reliable internet**HANDLED** Inference runs on an ordinary desktop CPU, about a minute per km²; results open in a 3-D viewer that ships as one offline HTML file.**OPEN RISKS, AND WHAT ANSWERS EACH****No Indian ground truth exists yet**Over India we can only compare against another model, Google Open Buildings, so we report agreement, not accuracy. **Answered by** SAC stereo or DGPS control points in the pilot (below).**Tall buildings compress toward the mean**Open training data holds few towers (tallest 83 m), so they are under-called. **Partly answered by** ISRO's GAMUS (it moved the gap significantly) and a control-point fit; fully, only by tall-building LiDAR.**DEPLOYMENT PLAN****1 Pilot**

Open Bhoonidhi CartoDEM and LISS-4 tiles; per-terrain metrics.

2 Validate

SAC Cartosat stereo or DGPS turns the Sikkim check into accuracy.

3 Integrate

Container behind an internal API; the viewer opens from disk.

4 Scale

Batch the archive: every scene gains a height layer.

COST ESTIMATE**Licences** ₹0 — every dependency open source**Training** one consumer RTX 3060 GPU**Compute** ~1 min per km² on a desktop CPU***Hosting** one two-core CPU server, live today**Imagery** already flown and already paid for**TESTING PLAN****Done** region-disjoint LiDAR validation**Done** second city, with a 338 m buffer**Done** ISRO's 0.6 m, simulated, on the shipped model**Next** held-out test split, scored once**Next** Indian truth: SAC stereo or DGPS*Derived: 5.0 s per 1024 px tile (0.094 km²), single pass, measured on a desktop CPU.

DISASTER USE CASE – BUILT, ON INDIAN TERRAIN



Raise the water and the viewer counts ground under water and buildings flooded, and flags the ones too close to call from its own sigma. Ground from Copernicus GLO-30; a flat-water model, labelled as one. Imagery © Maxar, CC BY-NC 4.0.

WHO IT SERVES



DISASTER RESPONSE

Post-event surfaces without waiting for a stereo pair.



ISRO AND SAC

Height for the single-view Cartosat archive.



URBAN PLANNING

Building heights for FSI and shadow checks (AMRUT).



FORESTRY

Canopy height proxies for biomass and encroachment.



TELECOM ROLLOUT

Line-of-sight and tower siting without a LiDAR survey.

0 new flights

input is imagery already flown and paid for

1 image

no stereo pair, no revisit wait

1 min / km²

on a desktop CPU, single pass*

0 licence fees

any state agency can adopt it

BEFORE AND AFTER, FOR THE PEOPLE WHO USE IT

	TODAY	WITH DEPTHWIZARD
SOCIAL Disaster response	Wait for a stereo pair or a survey flight.	A surface from one post-event image — about a minute per tile on a CPU web host.
SOCIAL Trust in the number	One height, no error bar.	Height plus a per-pixel sigma that ranks where it is likely wrong.
ECONOMIC The archive	Decades of single-view Cartosat with no height.	Every archived scene can gain a height layer, with no new tasking.
ECONOMIC Municipal bodies	Building heights need a survey budget.	Free and open; 36.8 MB runs offline on a laptop CPU.
ENVIRONMENTAL Survey flights	New LiDAR or stereo sorties to fly.	None: it reuses imagery already flown.
ENVIRONMENTAL Forest cover	Canopy height needs airborne LiDAR.	A canopy-height proxy from optical imagery alone.

Scales by batch, on CPUs agencies already own: the whole archive, not one scene. *Derived: 5.0 s per 1024 px tile (0.094 km²) measured on a desktop CPU.

RESEARCH THAT CHANGED WHAT WE BUILT

Depth Anything V2 Yang et al., NeurIPS 2024. The backbone we fine-tune.

HTC-DC Net Chen et al., TGRS 2023. Head-tail cut, distribution constraints.

Depth Any Canopy Rege Cambrin et al., ECCV-W 2024. The recipe for aerial height.

Beta-NLL Seitzer et al., ICLR 2022. Keeps the mean head learning.

IM2HEIGHT, TSE-Net Mou & Zhu 2018; Chen & Zhu 2025, both arXiv. Prior single-view height.

DATA AND REFERENCE SURFACES

DFC2019 Track 1 US3D at 0.3 m GSD with airborne LiDAR truth.

GAMUS ISRO's recommended set. 6,204 tiles, 0.33 m, DC and Philadelphia.

Copernicus GLO-30 Free 30 m global DEM; the absolute metric anchor.

GlobalBuildingAtlas Published 5.9 m RMSE over Asia — the external bar.

Google Open Buildings A cross-check over India, never used as truth.

Maxar Open Data Sikkim imagery, CC BY-NC 4.0, © Maxar Technologies.

TRIED, MEASURED, REJECTED

Ordinal / binned head All heads land at a 0.43–0.49 slope. No gain.

Shadow photogrammetry Oracle shadows reach r 0.503; our earlier net, 0.787.

LDS tail reweighting Pixels above 20 m are 7.5% but carry 78.8% of error.

Two-model ensemble Error correlation 0.925 — averaging buys nothing.

Global de-compression A rescale moves error, it does not remove it.

PROBES WE RAN, AND WHAT EACH CLOSED

01 Tall buildings Data, not capacity. Open 0.3 m sets hold few towers.

03 Guided filtering Squaring off roofs made every metric worse.

04 Where detail went Detail follows the ground area one token covers.

05b ISRO's 0.6 m Shipped model, 80 tiles: +15% per building after auto-zoom.

06 Height compression Measured: $0.473 \times \text{truth} + 2.66 \text{ m}$.

HOW WE COMPARE

PARAMETER	STEREO OR LIDAR SURVEY	PUBLISHED SINGLE-VIEW	DEPTHWIZARD
One archived image	✗	✓	✓
Absolute metric height	✓	—	✓
Per-pixel uncertainty	✗	✗	✓
Scored per terrain	—	✗	✓
Runs with no GPU	—	✗	✓
3D viewer shipped	✓	✗	✓
Failure modes published	—	—	✓

✓ yes — partial ✗ no

Two rows the alternatives win. A stereo survey gives absolute height directly and ships mature viewers — the table is only useful to a panel if the rows we lose are still in it.

EVIDENCE BEHIND EVERY NUMBER

Slide 7 is our preliminary work. Each figure on it regenerates from the evidence pack in the repository, and the documentation site walks through data, training, evaluation and failures in full.



Scan: pitch video

Live project5.zaidansari.tech
Docs project5.zaidansari.tech/documentation
Evidence github.com/.../sih2026-depthwizard/docs/evidence-pack.md

PRELIMINARY WORK

3.464 m

per-building RMSE vs airborne LiDAR;
5.9 m published for Asia

6.008 m

whole-tile RMSE on 80 held-out
tiles from unseen regions

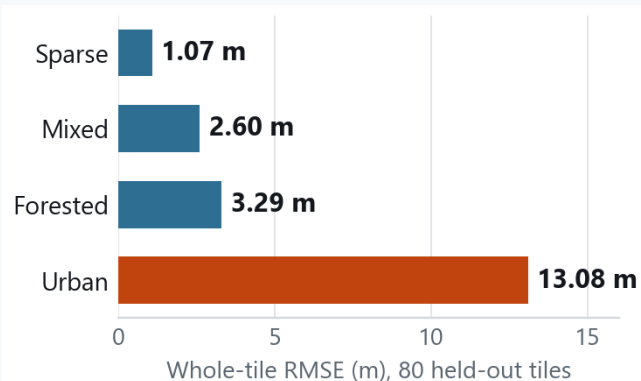
0.063

calibration error of the per-pixel
uncertainty (ECE)

+15 %

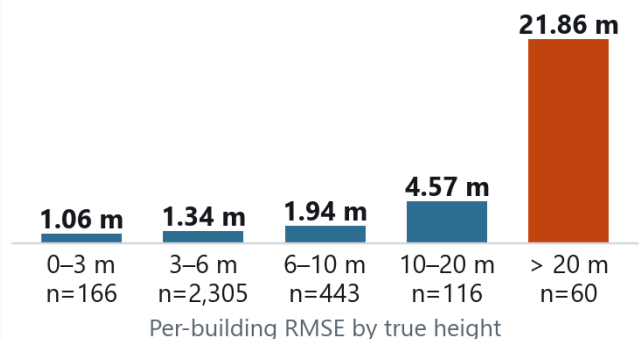
per-building error at ISRO's 0.6 m,
after auto-zoom (80 tiles)

ACCURACY ACROSS THE FOUR LANDSCAPES ISRO NAMES



Three of four classes at or under 3.3 m, correlation never below +0.77. Urban is one specific thing: the buildings above 20 m.
Hilly: DFC2019 has no hills, so Sikkim is a cross-check, not a score.

WHERE THE ERROR LIVES



77% of squared error comes from 1.9% of buildings. Under 10 m, 94% of the stock, we are at 1.4 m. **Above 20 m we under-call by 15.6 m;** we pre-registered -13 m and missed it.

ISRO'S 0.6 M CARTOSAT-2S, SIMULATED ON THE SHIPPED MODEL

INPUT	PER BUILDING	BUILDINGS ≤ 20 M	WHOLE TILE
0.3 m, native	3.464 m	1.667 m	6.008 m
0.6 m, untreated	4.185 m	2.090 m	6.738 m
0.6 m, auto-zoom	3.999 m	2.095 m	6.494 m

Auto-zoom cuts the 0.6 m penalty from +21% to +15%, all on tall buildings; under 20 m it does not help. We had pre-registered ≤ 10% and missed it. 0.6 m is simulated by downsampling, so this is the resolution gap, not the sensor gap.

BUILT AND WORKING TODAY

- ✓ **Elevation extraction**
fine-tuned DA-V2 with a sigma head
- ✓ **Scale calibration**
GLO-30 absolute DSM; GCP power-law fit
- ✓ **Visualisation**
three.js flythrough, measure, compare
- ✓ **Live web app + API**
upload a GeoTIFF, download the DSM
- ✓ **Flood tool**
water level over Indian terrain
- ✓ **Offline + CPU**
one HTML viewer; ONNX int8, no GPU

HOW THESE WERE MEASURED

80 whole held-out US tiles, 3,090 buildings, region-disjoint from training, against airborne LiDAR; no trained model has seen the test split. Full method: project5.zaidansari.tech/documentation



Scan: documentation